Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 06

Subject: BSC102/AP **Date: 09/11/2026**
Max Marks: 10 **Deadline: 20/11/2026**

CO6: Learner will be able to CLASSIFY semiconductors and EXPLAIN variation of Fermi level with temperature and doping concentration.

Q. No.	Questions	Marks	CO	CL
1	Calculate the probabilities for an electronic state to be occupied at 20°C, if the energy of these states lies 0.11 eV above and 0.11 eV below the Fermi level.	2	6	3
2	In a solid, consider the energy level lying 0.01 eV below Fermi level. What is the probability of this level not being occupied by an electron?	2	6	3
3	The intrinsic carrier density is 1.5×10^{16} m^{-3}. If the mobility of the electron and hole are 0.13 and 0.05 $m^2 V^{-1} s^{-1}$, calculate the conductivity.	2	6	2
4	The Intrinsic carrier density at room temperature in Ge is 2.37×10^{19} m^{-3} if the electron and hole mobilities are 0.38 and 0.18 $m^2 V^{-1} s^{-1}$ respectively, calculate the resistivity.	2	6	3
5	In a P-type germanium, $n_i = 2.1 \times 10^{19}$ m^{-3}, density of boron 4.5×10^{23} atoms/m^3. The electron and hole mobility are 0.4 and 0.2 $m^2 V^{-1} s^{-1}$ respectively. What is its conductivity before and after addition of boron atoms?	2	6	3